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Course: Data Analytics & Data Science
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SECTION A: Data fundamentals

① Database? Define, Why its essential. 2 examples

Its a collection of data, stored electronically so it can be easily accessed, updated and managed

Why is database important: Centralized storage allows consistent, reliable, effecient data across systems.

Example:

- Hospital patient record: A database that holds all patient's demographic, Medical history, Medication and lab results.
- A bank database that keeps customer accounts, balances & trans

② Data warehouse, How it differs from regular database Why would a company use data warehouse

A data warehouse is a centralized system used to store large amounts of data from diffent sources for analysis and reporting

Difference from a regular database:

- Database handles daily operations (transactions) shopnite transactions at the till.
- Data warehouses are used for analysis and decision making. ~~All the databases are combined~~

Why companies use it:

- To combine data from multiple systems for accurate,

Consistent reporting, historical trend analysis, KPI dashboard and cross functional insights without impacting transactional systems.

3. What is data lake? Explain concept and describe how it differs from a DW. When would a business partner prefer data lake over a DW?

A Data lake is a storage system that holds raw data in its original format (structured, semi-structured, & unstructured)

Difference from a data warehouse
- Data format: Lake stores diverse data types (Images etc)
- Schema handling: Lake uses Schema-on-read (more flexible)
- lake supports data science, ML, etc.

When business prefer a data lake:
- When dealing with large volume of diverse data
- When flexibility is needed.

4. What is Data Mart? How does it relate to a data warehouse and who typically uses it?

A data Mart is a smaller part of a data warehouse made for a specific team

Relatation to Datawarehouse
- It comes from a data warehouse.
Who uses it?
- Teams like sales, marketing or HR

5 Explain the Difference between structured, Semi-structured, and unstructured Data. Provide one example of each.

- Structured Data : Organized (tables)
- Semi-Structured data : partly organized (JSON)
- unstructured Data : Not organized (like video or images)

6. What is data modeling? Why it is important when designing a system that works with data? describe at least 2 types of data models

- Data modeling is planning how the data will be stored and connected

- Why important
* Makes data easy to use
* Improves system performance

- Types :
* Conceptual Model (basic idea)
* Logical model (more detail design)

7. What is data mining? How is it used in business? Provide 2 real-world examples of data mining in action.

Data mining meas finding useful patterns in large data

Examples :
- shops studying what customers buy
- Banks detecting fraud

8. What is a DBMS? Name, briefly describe 4 popular DBMS platforms and explain a suitable use case for each

A DBMS is a software used to create and manage databases.

Example:
* MySQL – used for websites
* PostgreSQL – for advanced data work
* Oracle – for big companies
* MongoDB – for flexible data

9. What is ETL? Explain each step and describe why this process is important in data analytics.

ETL means Extract, Transform, load
Extract – collect data
Tranform – Clean data
Load – Store data

Why its impart? Makes data ready for analysis.

10. What are common data storage formats used in analytics? Briefly describe each and state when you would use one over another.

* CSV – simple table format
* JSON – flexible format
* Parquet – good for big data
* Avro – fast and compact

References:

* IBM (2023) website. What is data science
* Google (2024) Machine learning crash course
* Open AI (2024) AI Overview
* Microsoft (2024)

SECTION B: AI & ML

11. What is AI? Define it and explain the difference between Narrow AI (weak AI) and general AI (Strong AI). Provide one example of each.

- AI is when machines can do tasks that need human thinking.
* Narrow AI: does one task
* General AI: can do many tasks (not fully developed yet)

12. What is machine learning? How does it differ from traditional programming? Describe the 3 main types of Machine learning: Supervised, unsupervised, and Reinforcement learing.

Machine learing is when computers learn from data instead of being programmed step by step.

Types:
* supervised - uses labeled data
* unsupervised - finds pattens
* Reinformed - learns from rewards

13. What is deep learning? How does it relate to ML and AI? What are neural networks and how do they work?

Deep learning is a type of machine learning that uses neural networks

Neural networks:
* Works like the human brain
* Process data in layers

14. What is Natural language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP helps computers understand human language.

Example:
* Chatbots
* Voice assistants
* Translation apps.

15. What is a large language Model (LLM)? How is it different from a standard machine learning model? Name two well-known LLM's and explain what they are used for.

An LLM is a large AI Model trained to understand and generate text

Example: - ChatGPT - text
- DALL-E - images
- Midjourney - art

17. What is a training dataset? Why is the quality of training data so important when building an AI model? What would go wrong if the training data is biased or incorrect?

Training dataset is the data used to train AI

why important:
- Good data = good results
- Bad data = wrong results

18. What is computer vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision

Computer vision allows machines to understand images

Example:
- face recognition
- self-driving cars

References:
- AWS website
- Google 2024 machine learning crash course
- OpenAI 2024

Section C: APIs & Tools & Developer Concepts

19. What is an API? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

An API lets different systems talk to ~~about~~ each other.

Example: like a waiter between you and the kitchen

20. What is an API key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

21. What is a Rest API? What does REST stand for and what are the main HTTP methods used in REST APIs? Briefly describe each method.

Rest is a way to use API's

Methods: * GET – get data
* POST – Send data
* PUT – update data
* DELETE – remove data

22.

JSON is a simple format for sharing data

Example: </> JSON

```
{ "name": "John",
  "age": 25 }
```

23. Prompt Engineering

It means writing clear instructions for AI

Example:

* Poor: "write about data"
* Better: "Explain data analytics in 200 words"

24.

It applies a function to each item in a list.

Example:

```
</> python

numbers = [1, 2, 3]
result = list(map(lambda x: x * 2, numbers))
```

~~25~~ 26

Version control tracks changes in code

* Git - tool used
* commit - saved change
* Branch - different Version

~~25~~ Version control tracks changes in code

25.
Cloud computing is using the internet to store and use data

- Iaas – Infrastructure
- Paas – Platform
- Saas – Software

27.

Jupiter notebook is a tool to write and run code.

Why used:
- Shows code and results together

28.
Python is a popular programing language

Libraries

- Pandas – data analysis
- Numpy – Calculations
- Matplotlib – Charts
- Scikit-learn – Machine learning

29.

Reference:

- Google
- AWS
- Oracle

Section D: Data ethics, governance & careers

29.

Data Privacy is keeping personal data safe

GDPR: A law in europe that protects data

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Data ethics is using data in a fair and honest way

Example

* Selling data without permision
* Biased AI
* Misleading results

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Data governance is managing data properly

Includes: - Quality
- Security
- Rules

32

Biased AI gives unfair results
Example: Hiring systems favoring certain people.

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Role	focus	Skills	Tools
Data Analyst	Data insights	SQL, Excel	Power BI
Data Scientist	Predictions	Python, ML	Scikit-learn
Data engineer	Data systems	coding	Spark
AI engineer	AI systems	ML, deep learning	Tensorflow

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Business Intelligence is tools that help businesses understand data

Example:
power BI
Tableau

Reference

European Comission GDPR overview
IBM
Microsof